

Notes from meeting w Mick:

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- all supplies in dry box c7
- Some small notations on large excel document to aid in understanding run coding, can share my copy with you directly
- SPTS_DRIE_recipe.xlsx includes proposed changes in blue for NEW runs
- Etch time to 2.5-3.5 seconds (specify on tool, linear ramp), increased from previous time
- Process pressure increased for deposition, decreased for etch (previous 40, 30, 30; suggested 50, 20, 20)
- At this time no recommendations to changing platen power, may want to change in the future
- Document shows parameter of the tool needing ran
- Larger holdup: Grace unsure of tool viability to attempt these pressures, likely need to wait until tool is in better shape. Process flow could be nailed down ahead of time, getting wafers ready. Could try 40, 30, 30 just with increased etch time preliminarily.
- Need to figure out photolithography. Currently is an etch mask 1um, 3um, 5um trenches, 5um square pillars. Suggesting to use MLA 150 direct write instead for more flexibility in wafer design. Samyama can use Claude to make own mask. Only want to do a few wafers, would be faster. Jonah and Labib are currently working on similar MLA masks - have good starting point for photo settings. MLA more likely to have better photo. Important for this project to have well defined features.
- For 4um oxide, should only need ~4-8 um of resist
- Before DRIE runs, need to test etching properly, ICP-F is down, ICP-7 recommended by Grace for oxide etching, recipe unproven currently. Tests need to be done with that first. Mick says Keyence would likely be good for examining how that recipe etches wafers. Could likely re-use wafers for oxide-etch test. Will reveal "perfect" amount of time to etch through oxide without through top of photoresist. Will show if we need more photoresist or need to tweak recipe. Also test for photoresist reticulation.
- For wafer images, C=center, E=East, S=South, other directions, etc. relative for the North at major flat.
- Next Steps:
 - Determine if any of the old wafers should be retained (because Fred is planning to use some for the ALD conformality project)
 - obtain 4um oxide wafers
 - spin coat 7um of resist
 - make new design
 - do photo on the new design with MLA (Jonah/Labib may need to figure out)
 - After figuring out photo, test out Grace's ICP-7 recipe
 - Wafers can be re-used for testing
 - use Keyence (maybe Nanospec) to see how that recipe is working (SEM if necessary)
 - After done - can do DRIE runs, up to us how to proceed with full proposed or partial proposed recipe (same pressures, different etch time), or same recipe as before (to test against Mick's previous results), and # of loops for 3rd loop set
 - Then SEM for metrology analysis

- On batch 3, 10 wafers already have oxide on them, 500um wafers, 6um of oxide, could be re-used for something else, but 6um of oxide is generally too much for these applications. Recommended to shave down wafers with etch RIE - can do Filmetrics before and after - get uniform 4um mask, also vapor HF etcher, or use BOE to strip entirely then re-deposit. Wafers will be in C7 dry box in labeled cassette. Should have label, DRIE Project